

Chapter 4. Lagrange's Theorem

4.1 Cosets



In this video, we will some important properties of cosets.

Some important properties of cosets

Let $a \in G$ and H a subgroup of G . Then:

1. $H1 = H = 1H$, that is, H is a right and left coset of itself.
2. It is obvious that $a \in Ha$ and $a \in aH$ since $a = 1 \cdot a = a \cdot 1$ and $1 \in H$.
3. If G is abelian, then $Ha = aH$ for all $a \in G$ and all subgroups H of G .
4. 3 might not be true if G is not abelian.

Take $G = S_3$ and $H = \{e, (1\ 3)\}$, $a = (1\ 2)$
 $(1\ 2)H \neq H(1\ 2)$.

5. If $a \in Hb$, then $Ha = Hb$. ($b \in G$).

$$\underline{Ha} \overset{?}{\subseteq} Hb$$

Since $a \in Hb$, then $a = hb$ for some $h \in H$.

If $x \in Ha$, then $x = h_1 a$ for some $h_1 \in H$.

Our goal is to show that $x \in Hb$.

$$\underline{x = h_1 a = h_1 (hb) = \underbrace{(h_1 h)}_H b \in Hb}$$

To show that $Hb \subseteq Ha$, proceed in the same way using now that $b = h^{-1}a$.

6. $Ha = Hb$ if and only if $ab^{-1} \in H$.

7. $Ha = H$ if and only if $a \in H$. (It follows from 6 by taking $b = 1$)

In additive notation 6 and 7 become

$$H + a = H + b \Leftrightarrow a - b \in H; \quad H + a = H \Leftrightarrow a \in H$$

For instance, $6\mathbb{Z} + 7 = 6\mathbb{Z} + 1$ since $7 - 1 = 6 \in 6\mathbb{Z}$.

Proof of 6. Suppose first that $ab^{-1} \in H$. Then $ab^{-1} = h$ for some $h \in H$. This implies that $\underline{a} = hb \in \underline{Hb}$. By property 5 we have that $Ha = Hb$.

Assume now that $Ha = Hb$ and let us show that $ab^{-1} \in H$.

If $Ha = Hb$, then $\underline{a} \in Ha = \underline{Hb}$ and so $a = hb$ for some $h \in H$. But then $h = ab^{-1}$, which implies that $ab^{-1} \in H$.

8. There exists a bijective map from H to Ha . In particular, if H is finite, then any coset Ha has the same number of elements as H and therefore all the cosets have the same number of elements.

Define $\varphi: H \longrightarrow Ha$
$$h \longmapsto \varphi(h) = ha$$

φ is surjective: Any element x of Ha is of the form $x = h_1 a$ for some $h_1 \in H$. So, $x = h_1 a = \varphi(h_1)$.

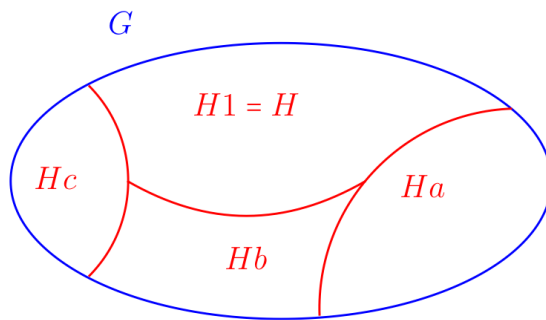
φ is injective: Suppose $\varphi(h_1) = \varphi(h_2)$, and let us show that $h_1 = h_2$.

$$\varphi(h_1) = \varphi(h_2) \Rightarrow h_1 a = h_2 a \Rightarrow$$

Cancellation
laws

$$\underline{h_1 = h_2} \longrightarrow$$

9. The family of all the cosets of H , as a ranges over G , is a partition of G . It means that $G = \cup_{a \in G} Ha$, and $Ha \neq Hb$ implies $Ha \cap Hb = \emptyset$.



We first prove that two cosets Ha and Hb (where a and b are in G) are either equal or disjoint. If $Ha \cap Hb = \emptyset$ (Ha and Hb are disjoint), then we are done. Otherwise, $Ha \cap Hb \neq \emptyset$, and take $x \in Ha \cap Hb$.

Then $x = h_1a = h_2b$, for some h_1 and h_2 in H .

$$h_1 a = h_2 b \Rightarrow \underline{a = (h_1^{-1} h_2) b} \in \underline{Hb}$$

By property (5) we obtain that

$$\underline{Ha = Hb}.$$

Clearly, $G = \bigcup_{a \in G} Ha$, because

for any $g \in G$ we have that

$$\underline{g \in Hg}.$$